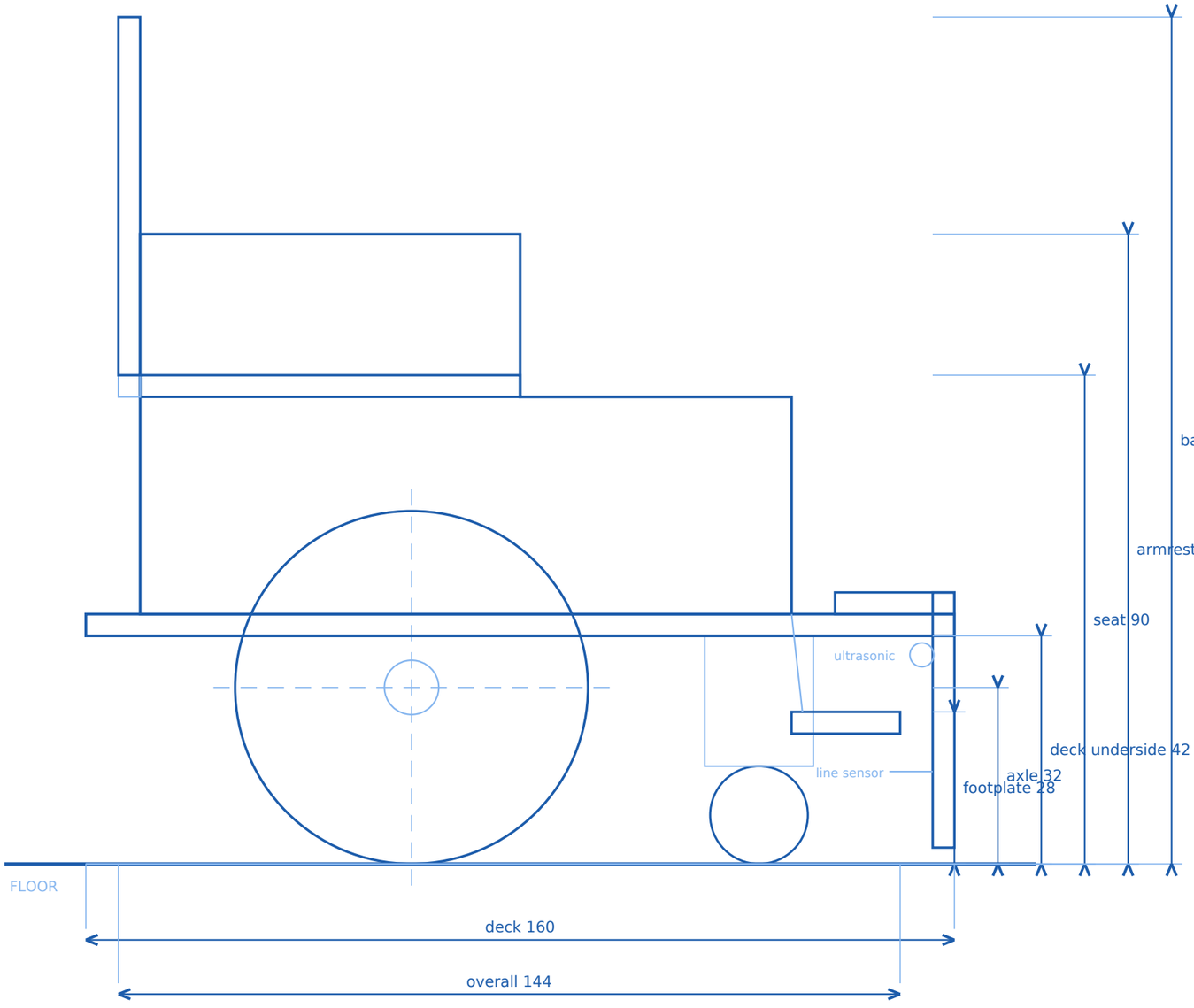
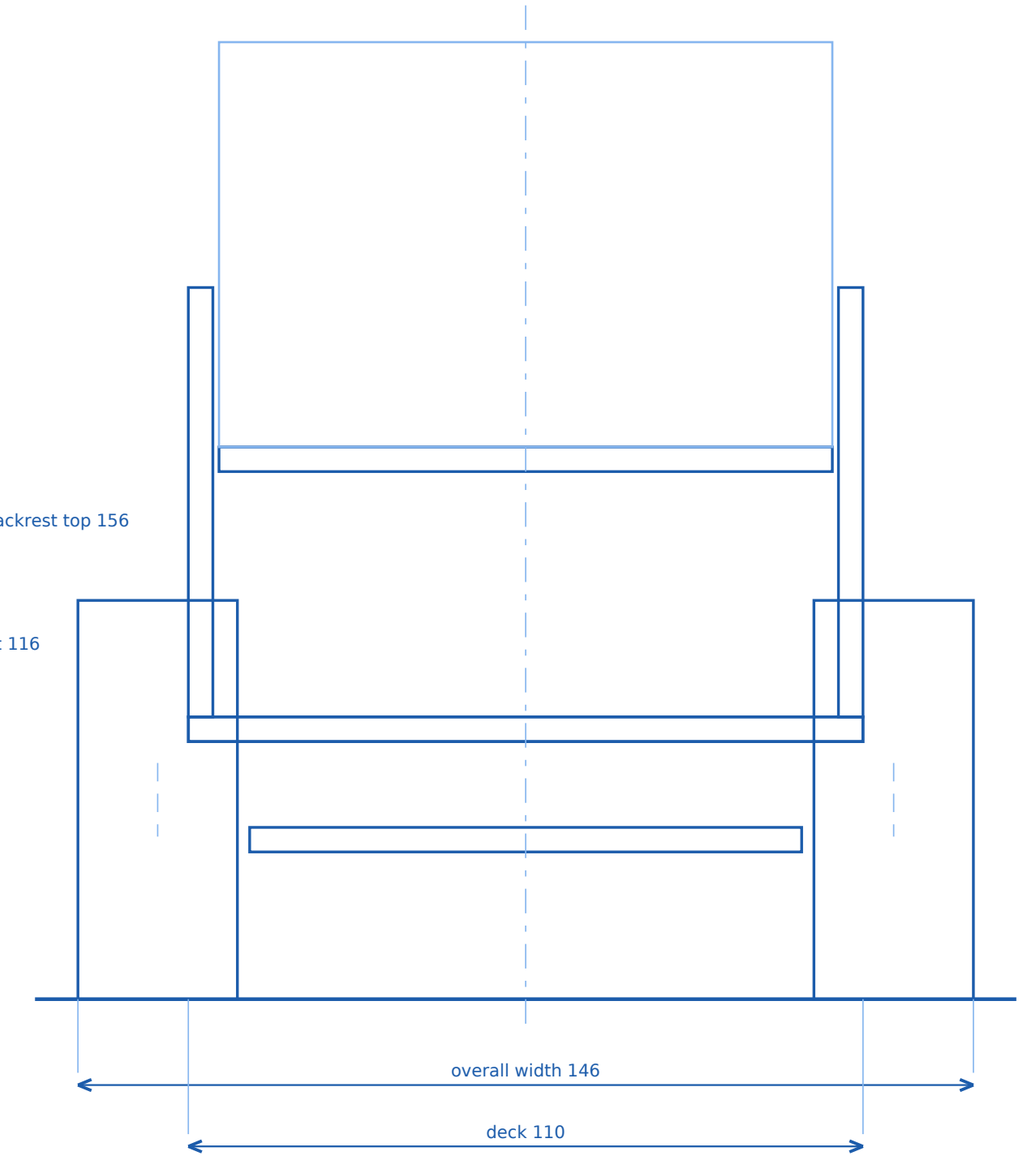


SIDE ELEVATION

looking from the chair's left; forward to the right



FRONT ELEVATION

backrest shown light, behind the seat

CRITICAL DIMENSIONS

Ride height is set by the wheel: 65 mm dia gives a 42 mm deck underside.
A different wheel changes everything above it. Edit WHEEL_DIA and re-run both generators.
Line sensor adjusts 9–25 mm above the floor, by sliding in its slots.
Set it on the real track surface, not on the bench.

ASSEMBLY

1 Castor and riser first. Check the deck sits level on the wheels before anything else.
2 Motors, zip-tied tight. A motor that can rotate in its mount makes the line following untunable, and you will blame the PID for a week.
3 Side panels with gussets inside the corners, square to the deck.
4 Seat cleats inside the panels, then the seat pan, then the backrest.
5 Sensor bracket last, screws left loose until the sensor height is set.

Dry-fit the whole body with tape before any glue. MDF and wood glue give you ten seconds.

SAARTHI — GUIDED ASSISTIVE CHAIR, SCALE DEMONSTRATOR

Scale	1:1 on A3. The model is 1:6 of a real wheelchair.
Material	4 mm MDF throughout, except the wheels and castor.
Units	Millimetres. Heights are above the floor.
Drive	2 × BO geared motor, 65 mm wheels, differential, castor at the front.
Mass	≈ 600 g loaded.
Drawn from	generate_chassis.py and generate_mdf_body_template.py